Sin U Choi
DSC 224 Final Project
06/09/2026

Comparing Cox Proportional Hazards and Random Survival Forest Models to Predict All-Cause Mortality in Prostate Cancer Survivors

Section 1: Problem and Dataset

Background

Prostate cancer is the most diagnosed cancer among American men[1], and while most diagnoses are not immediately lethal, long-term survival is shaped by both disease severity and modifiable lifestyle factors. Diet quality, especially adherence to a plant-based dietary pattern, has been increasingly linked to survival outcomes in this population. The Plant-Based Diet Index (PDI) is a validated composite dietary score ranging from 18 to 90, constructed from 18 food groups[2] derived from a food frequency questionnaire (FFQ). Higher PDI scores reflect greater adherence to a plant-based dietary pattern. While PDI has been linked to prostate cancer progression in CaPSURE, whether machine learning methods offer predictive gains over Cox regression for all-cause mortality (ACM) in this population has not been examined.[3]

Dataset Description

This project utilized data from the Cancer of the Prostate Strategic Urologic Research Endeavor (CaPSURE), a longitudinal, community-based prostate cancer registry.[4] The analytic cohort was drawn from the CaPSURE Diet and Lifestyle substudy, which enrolled men with localized disease (clinical T-stage ≤ T3a), a completed FFQ after diagnosis, known primary treatment, and total energy intake within plausible bounds (800-4,200 kcal/day). A complete-case restriction was applied, yielding a primary analytic sample of N = 1,813. The outcome of interest was ACM, ascertained through linkage to the National Death Index. Follow-up was defined from the date of FFQ completion to death or administrative censoring on December 31, 2020. A total of 693 deaths were observed (event rate 38.2%) over a median follow-up of 6.0 years.

Baseline Characteristics

The primary exposure was PDI analyzed as a continuous score. Thirteen covariates were included: age at FFQ completion, log-transformed total energy intake, years from diagnosis to FFQ, Gleason grade, PSA category, clinical T-stage, primary treatment modality, BMI, smoking status, race/ethnicity, walking pace, and diabetes status. Baseline characteristics of the full analytic cohort (N=1,856 prior to complete-case restriction) stratified by PDI quintile are presented in Table 1. Participants in higher PDI quintiles were modestly older, had lower BMI, and lower rates of diabetes compared to those in lower quintiles.

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Table 1. Baseline characteristics of 1,856 localized prostate cancer patients in CaPSURE, stratified by plant-based diet index (PDI) quintile. Data are median (IQR) for continuous variables and n (%) for categorical variables.

Characteristic	**Overall** N = 1,856[1]	**Q1(lowest)** N = 372[1]	**Q2** N = 371[1]	**Q3** N = 371[1]	**Q4** N = 371[1]	**Q5(highest)** N = 371[1]	**p-value**[2]
Age at diagnosis (years)	65 (60, 71)	64 (58, 70)	65 (59, 71)	65 (59, 70)	66 (61, 71)	67 (61, 72)	**0.003**
BMI (kg/m²)	26.9 (24.7, 29.6)	28.0 (25.8, 30.9)	26.9 (24.7, 29.8)	27.2 (25.1, 29.6)	26.4 (24.5, 28.8)	25.8 (23.7, 28.7)	**<0.001**
PSA at diagnosis (ng/mL)	5.9 (4.5, 8.6)	6.2 (4.5, 9.1)	5.9 (4.5, 9.5)	5.6 (4.5, 7.9)	5.7 (4.4, 8.2)	5.9 (4.5, 8.3)	0.2
Total energy intake (kcal/day)	1,931 (1,522, 2,362)	1,636 (1,264, 2,021)	1,753 (1,468, 2,179)	1,905 (1,507, 2,317)	2,024 (1,584, 2,463)	2,304 (1,947, 2,786)	**<0.001**
Alcohol intake (servings/day)	0.29 (0.00, 1.20)	0.43 (0.00, 1.43)	0.29 (0.00, 1.21)	0.21 (0.00, 1.14)	0.43 (0.00, 1.14)	0.29 (0.00, 1.07)	0.6
Wine intake (servings/day)	0.07 (0.00, 0.29)	0.00 (0.00, 0.18)	0.00 (0.00, 0.29)	0.00 (0.00, 0.21)	0.07 (0.00, 0.29)	0.07 (0.00, 0.50)	0.10
Race/ethnicity							0.15
White	1,781 (96.0%)	360 (96.8%)	355 (95.7%)	350 (94.3%)	354 (95.4%)	362 (97.6%)	
Black/African American	42 (2.3%)	8 (2.2%)	5 (1.3%)	14 (3.8%)	10 (2.7%)	5 (1.3%)	
Other	33 (1.8%)	4 (1.1%)	11 (3.0%)	7 (1.9%)	7 (1.9%)	4 (1.1%)	

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Smoking status							0.082
Never	765 (41.2%)	140 (37.6%)	159 (42.9%)	139 (37.5%)	165 (44.5%)	162 (43.7%)	
Former	977 (52.6%)	201 (54.0%)	189 (50.9%)	204 (55.0%)	193 (52.0%)	190 (51.2%)	
Current	114 (6.1%)	31 (8.3%)	23 (6.2%)	28 (7.5%)	13 (3.5%)	19 (5.1%)	
Walking pace							0.071
Normal	899 (49.6%)	195 (54.0%)	181 (49.9%)	159 (44.2%)	185 (50.4%)	179 (49.4%)	
Easy	297 (16.4%)	59 (16.3%)	69 (19.0%)	66 (18.3%)	56 (15.3%)	47 (13.0%)	
Unable	27 (1.5%)	7 (1.9%)	8 (2.2%)	8 (2.2%)	1 (0.3%)	3 (0.8%)	
Brisk	523 (28.8%)	90 (24.9%)	93 (25.6%)	112 (31.1%)	113 (30.8%)	115 (31.8%)	
Fast	67 (3.7%)	10 (2.8%)	12 (3.3%)	15 (4.2%)	12 (3.3%)	18 (5.0%)	
Family history of prostate cancer	344 (18.5%)	58 (15.6%)	78 (21.0%)	73 (19.7%)	61 (16.4%)	74 (19.9%)	0.2

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Characteristic	Overall N = 1,856[1]	Q1(lowest) N = 372[1]	Q2 N = 371[1]	Q3 N = 371[1]	Q4 N = 371[1]	Q5(highest) N = 371[1]	p-value[2]
Diabetes	206 (11.1%)	47 (12.6%)	57 (15.4%)	41 (11.1%)	34 (9.2%)	27 (7.3%)	**0.006**
Gleason score at diagnosis							**0.016**
<7	1,237 (66.6%)	242 (65.1%)	245 (66.0%)	256 (69.0%)	263 (70.9%)	231 (62.3%)	
7	475 (25.6%)	104 (28.0%)	108 (29.1%)	83 (22.4%)	81 (21.8%)	99 (26.7%)	
>7	144 (7.8%)	26 (7.0%)	18 (4.9%)	32 (8.6%)	27 (7.3%)	41 (11.1%)	
Clinical T-stage at diagnosis							**0.048**
T1	976 (52.6%)	209 (56.2%)	205 (55.3%)	180 (48.5%)	203 (54.7%)	179 (48.2%)	
T2	855 (46.1%)	156 (41.9%)	165 (44.5%)	188 (50.7%)	161 (43.4%)	185 (49.9%)	
T3a	25 (1.3%)	7 (1.9%)	1 (0.3%)	3 (0.8%)	7 (1.9%)	7 (1.9%)	
Primary treatment							0.9
Radical Prostatectomy	1,117 (60.2%)	233 (62.6%)	221 (59.6%)	221 (59.6%)	223 (60.1%)	219 (59.0%)	

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Radiation Therapy	431 (23.2%)	85 (22.8%)	93 (25.1%)	78 (21.0%)	89 (24.0%)	86 (23.2%)	
Androgen Deprivation Therapy	130 (7.0%)	26 (7.0%)	21 (5.7%)	31 (8.4%)	25 (6.7%)	27 (7.3%)	
Watchful Waiting/Other	178 (9.6%)	28 (7.5%)	36 (9.7%)	41 (11.1%)	34 (9.2%)	39 (10.5%)	
Multivitamin use							0.2
Never	380 (20.9%)	82 (22.4%)	76 (21.1%)	76 (21.0%)	69 (18.9%)	77 (20.9%)	
Former	304 (16.7%)	72 (19.7%)	63 (17.5%)	66 (18.2%)	58 (15.9%)	45 (12.2%)	
Current	1,138 (62.5%)	212 (57.9%)	221 (61.4%)	220 (60.8%)	238 (65.2%)	247 (66.9%)	
Years from diagnosis to FFQ	2.6 (1.4, 4.7)	2.8 (1.5, 5.1)	2.5 (1.3, 4.7)	2.4 (1.3, 4.3)	2.5 (1.4, 4.4)	2.7 (1.3, 5.2)	0.15

[1] Median (Q1, Q3); n (%)

[2] Kruskal-Wallis rank sum test; Pearson's Chi-squared test

Note: Age at FFQ completion used as covariate in multivariable Cox models.

Missing: walking pace (n=43); multivitamin use (n=34).

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Section 2: The Random Survival Forest Algorithm

Cox proportional hazards regression is the standard method for survival analysis but rests on a critical assumption, that the hazard ratio between any two individuals must remain constant over time. Formal evaluation of this assumption in the primary Cox model revealed violations for 3 covariates: treatment modality, age at FFQ completion, and years from diagnosis to FFQ. Random Survival Forest (RSF) was selected as a non-parametric machine learning alternative that naturally accommodates time-varying covariate effects, nonlinear relationships, and variable interactions without requiring pre-specification.[5]

The Mechanism of RSF

RSF is an ensemble method that aggregates predictions across B survival trees, each built on a bootstrap sample of the training data. At each node of each tree, a random subset of $m = \lfloor\sqrt{p}\rfloor$ covariates are considered as candidates for splitting, where p is the total number of features. Randomly restricting which features can be split at each node forces trees to decorrelate and reduces overall variance.[6] The optimal split maximizes the log-rank statistic between the two resulting child nodes

$$L = \frac{\sum_{t_j \in D}\left(d_{1j} - e_{1j}\right)}{\sqrt{\sum_{t_j \in D} V_j}}$$

Where d_{1j} is the observed number of deaths in child node 1 at time t, $e_{1j} = n_{1j} \cdot \frac{d_j}{n_j}$ is the expected number under the null hypothesis of equal survival between child nodes, V_j is the hypergeometric variance at time t_j, and $D$is the set of all unique death times. This criterion is mathematically equivalent to the log-rank test to compare Kaplan Meier curves.

Splitting continues recursively until each terminal node contains a minimum number of observations (set to 5 in this analysis to prevent overfitting). Within each terminal node, the Nelson-Aalen estimator is then applied to estimate the cumulative hazard function[7,8]:

$$\widehat{\mathrm{H}}(\mathrm{t}) = \sum_{t_j \leq t} \frac{d_j}{n_j}$$

where d_j is the number of deaths at time t_j and n_j is the number of participants at risk at time t_j.

For a new patient, each of the $B = 300$ trees produce an estimated cumulative hazard function based on the terminal node the patient falls into. These are averaged across all trees to obtain the ensemble prediction:

$$\widehat{H}_{ensemble}(t|x) = \frac{1}{B}\sum_{b=1}^{B}\widehat{H}_b(t|x)$$

Where b is the index of the specific tree at given time of the patient with covariate x.

Predicted risk scores used for computing the C-index is computed as the ensemble cumulative hazard summed over all observed event times:

$$\widehat{\Lambda}(x) = \sum_{t_j \in D}\widehat{H}_{ensemble}\,(t_j|x)$$

Higher values of $\widehat{\Lambda}(x)$indicate greater predicted mortality risk.

Model calibration was assessed using the Integrated Brier Score (IBS), which measures the mean squared difference between predicted survival probabilities and observed outcomes across all timepoints:

$$IBS = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{w}\int_0^w[1(y_i > t) - \hat{s}(t|x_i)]^2dt$$

Where w is the maximum observed time, $1(y_i > t)$ is the observed survival status at time t, and $\hat{s}(t|x_i)$ is the predicted survival probability. Lower IBS indicates better calibration and a null model scores 0.25.

Assumptions

RSF makes several assumptions. First, observations are assumed to be independent which is satisfied given that CaPSURE enrolls one observation per patient. Second, survival time and censoring time are assumed to be conditionally independent given the covariates. Third, RSF assumes that each tree contributes equally to the ensemble prediction regardless of individual tree performance. Unlike Cox, RSF makes no assumption about the functional form of the relationship between covariates and the hazard, and no assumption of proportionality which is the primary advantage in this application.

RSF Optimal Condition on Performance

RSF tends to perform best when covariate-outcome relationships are nonlinear or involve interactions, when the proportional hazards assumption is violated, and when sample sizes are large enough to support deep tree growth. Log-rank RSF performs competitively for point prediction at low censoring rates but consistently below average for survival function estimation across most settings, particularly at higher censoring rates.[9] With 61.8% censoring in the CaPSURE dataset, the limited observed events at each split reduces the log-rank test’s ability to detect meaningful survival differences, resulting in noisier trees and attenuated ensemble performance.

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Model Performance

All analyses were conducted in Python using the scikit-survival library.[10] Results from applying Cox proportional hazards regression and Random Survival Forest to the CaPSURE analytic cohort are presented in Table 2. The dataset was partitioned into a training set (80%, n = 1,450) and a test set (20%, n = 363) using stratified random splitting to preserve the event rate. Five-fold cross-validation was performed to obtain stable estimates of model discrimination, as single train/test splits are sensitive to random partitioning in modestly sized datasets. Cross-validation was applied to Cox PH to enable a fair comparison, since RSF inherently produces out-of-bag estimates during bootstrap sampling.

Cox PH achieved a test set C-index of 0.705 and a five-fold cross-validated C-index of 0.700+/- 0.027, consistent with the concordance statistic of 0.714 (from the primary Cox analysis on the full pre-restriction cohort, N = 1856). RSF achieved a test set of C-index of 0.698 and a cross-validated C-index of 0.691+/- 0.029. Per-fold C-index values tracked closely between models across all five folds, with one fold where RSF marginally exceeded Cox. This consistency suggests that performance differences reflect data characteristics rather than model-specific behavior.

Integrated Brier Score (IBS) was used to assess calibration and the result favored Cox PH (IBS = 0.111) over RSF (IBS = 0.116), where lower values indicate closer agreement between predicted survival probabilities and observed outcomes. Both scores are well below the null model benchmark of 0.25, confirming that both models learn real patterns rather than fitting noise. Both metrics indicated that Cox marginally outperformed RSF.

Table 2. Model discrimination and calibration for Cox PH and RSF applied to 1,813 localized prostate cancer survivors in CaPSURE.

Model	**Test C-index**	**CV-C-index(5-fold)**	**IBS**
Cox PH	0.705	0.700 ± 0.027	0.111
RSF (log-rank)	0.698	0.691 ± 0.029	0.116

C-index: Harrell's concordance index[11]; higher = better discrimination. IBS: Integrated Brier Score; lower = better calibration. CV: cross-validated.

Variable Importance

Permutation-based variable importance from RSF is presented in Figure 2. Age at FFQ completion was the dominant predictor by a substantial margin (importance = 0.103), followed by years from diagnosis to FFQ (0.023), BMI (0.010), and current smoking status (0.009). This is expected because older prostate cancer patients are more likely to die from age-related causes like cardiovascular disease than from cancer. PDI ranked 12th in importance (0.001), reflecting a small but present contribution consistent with the statistically significant association observed in the primary Cox analysis (HR = 0.983, 95% CI: 0.972-0.995). Traditional clinical cancer severity markers (Gleason grade, PSA category, and T-stage) ranked among the lower importance features, as cancer severity adds limited predictive signal for all-cause mortality.

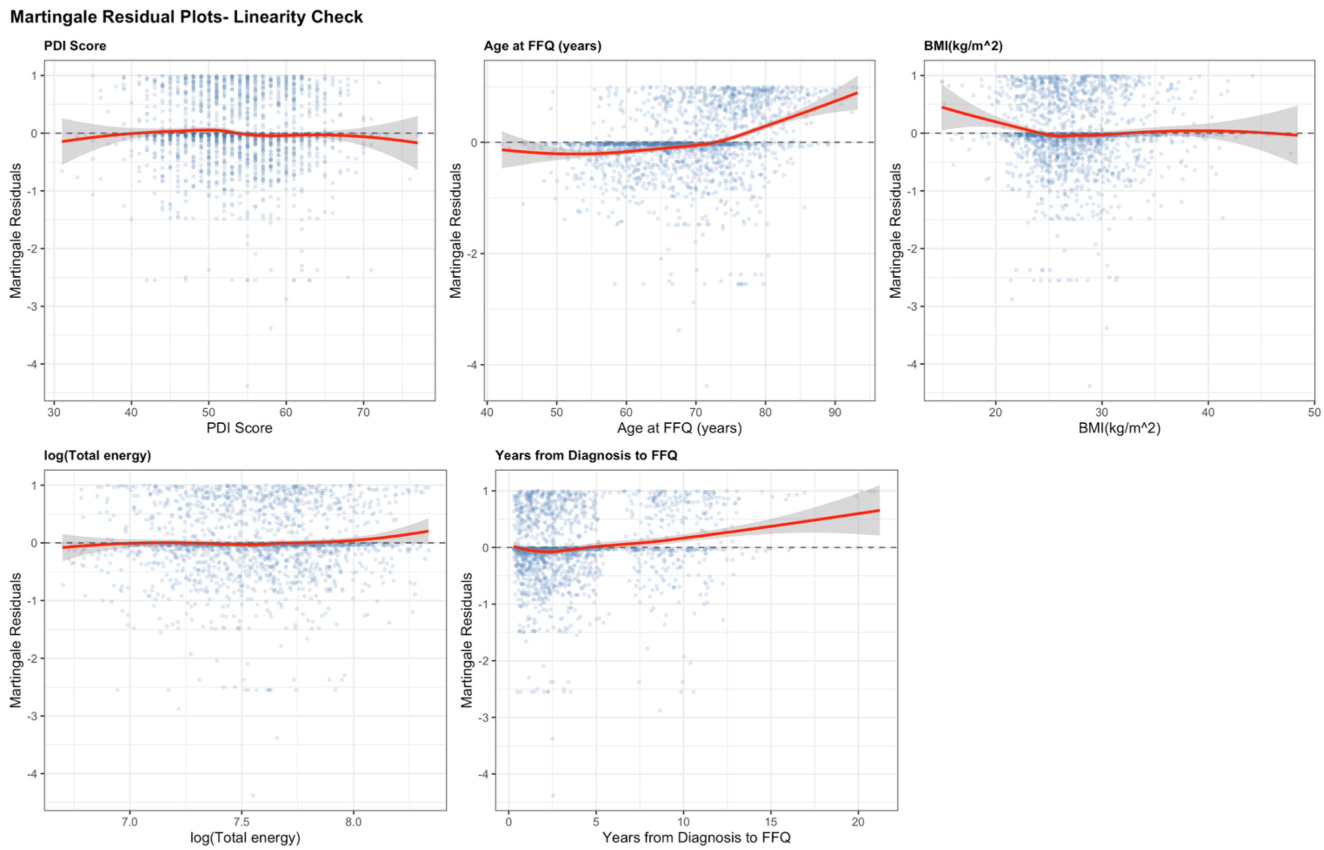

Figure 1. Martingale residual plots for continuous covariates in the fully adjusted Cox proportional hazards model (N = 1,813). Red line indicates LOESS smoother; dashed line at zero indicates linearity. PDI score demonstrates an approximately linear relationship with the log hazard, supporting the adequacy of the linear Cox specification and providing context for the comparable performance of Cox and RSF.

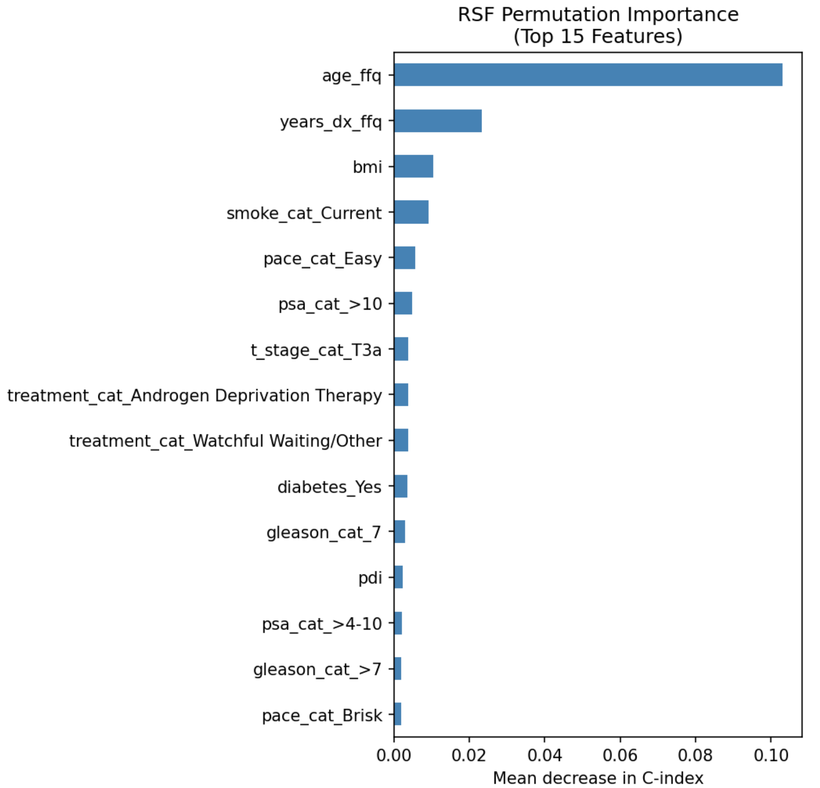

Figure 1. RSF permutation-based variable importance (mean decrease in C-index) for the top 15 dummy-expanded features across 13 predictors.

Discussion

Cox PH marginally outperformed RSF across both discrimination and calibration metrics. This is not surprising for two reasons. First, the PDI-ACM relationship is approximately linear (Martingale residual analysis) and additive after covariate adjustment, such that additional flexibility of RSF does not translate into meaningful predictive gains. Second, the high censoring rate (61.8%) likely attenuated RSF performance relative to its theoretical potential. Log-rank RSF performs consistently below average for survival function estimation at high censoring rates, and alternative forest methods such as RotSF[12] show stronger performance in information-scarce settings[9]. Future analyses with this dataset might consider these alternative splitting criteria.

Despite its marginal underperformance, RSF contributed meaningful insight through variable importance, confirming age and time-related variables as dominant predictors of ACM, consistent with the small but statistically significant PDI effect observed in the Cox model. RSF confirms the signal exists without assuming linearity or proportionality; Cox provides a hazard ratio that can be clinically interpreted. In prostate cancer survivorship research, where clinical translation matters as much as predictive accuracy, the two approaches work best together as complementary tools.

References

1. Siegel RL, Giaquinto AN, Jemal A. Cancer statistics, 2024. *CA Cancer J Clin*. 2024;74(1):12-49. doi:10.3322/caac.21820

2. Satija A, Bhupathiraju SN, Rimm EB, et al. Plant-Based Dietary Patterns and Incidence of Type 2 Diabetes in US Men and Women: Results from Three Prospective Cohort Studies. Moore SC, ed. *PLOS Med*. 2016;13(6):e1002039. doi:10.1371/journal.pmed.1002039

3. Liu VN, Van Blarigan EL, Zhang L, et al. Plant-Based Diets and Disease Progression in Men With Prostate Cancer. *JAMA Netw Open*. 2024;7(5):e249053. doi:10.1001/jamanetworkopen.2024.9053

4. Lubeck DP, Litwin MS, Henning JM, et al. The capsure database: a methodology for clinical practice and research in prostate cancer. *Urology*. 1996;48(5):773-777. doi:10.1016/S0090-4295(96)00226-9

5. Ishwaran H, Kogalur UB, Blackstone EH, Lauer MS. Random survival forests. *Ann Appl Stat*. 2008;2(3):841-860. doi:10.1214/08-AOAS169

6. Breiman L. Random Forests. *Mach Learn*. 2001;45(1):5-32. doi:10.1023/A:1010933404324

7. Nelson W. Theory and Applications of Hazard Plotting for Censored Failure Data. *Technometrics*. 1972;14(4):945-966. doi:10.1080/00401706.1972.10488991

8. Aalen O. Nonparametric Inference for a Family of Counting Processes. *Ann Stat*. 1978;6(4):701-726. doi:10.1214/aos/1176344247

9. Berkowitz M, Altman RM, Loughin TM. Random forests for survival data: which methods work best and under what conditions? *Int J Biostat*. 2024;20(2):315-345. doi:10.1515/ijb-2023-0056

10. Pölsterl S. scikit-survival: A Library for Time-to-Event Analysis Built on Top of scikit-learn. *J Mach Learn Res.* 2020;21(212). http://jmlr.org/papers/v21/20-729.html

11. Harrell FE, Lee KL, Mark DB. Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Stat Med*. 1996;15(4):361-387. doi:10.1002/(SICI)1097-0258(19960229)15:4<361::AID-SIM168>3.0.CO;2-4

12. Zhou L, Xu Q, Wang H. Rotation survival forest for right censored data. *PeerJ*. 2015;3:e1009. doi:10.7717/peerj.1009

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